

Confusion Matrix

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Scores and Calibration Intuition

Without opening the old lesson, explain one key idea from Scores and Calibration Intuition. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Nearest Neighbors

Without opening the old lesson, explain one key idea from Nearest Neighbors. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Dataset Cards

Without opening the old lesson, explain one key idea from Dataset Cards. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Explain the two error types for Confusion matrices. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Confusion matrices. Proposed explanation: “Positive means good”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Changed independent case

On 20 cases: true positives 5, false negatives 4, false positives 2, true negatives 9. Build the confusion matrix with actual class as rows and predicted class as columns. Compute accuracy.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Positive = spotted. TP=6, FN=2, FP=3, TN=9. Draw a matrix with actual labels in rows and predictions in columns. Compute total and accuracy.

6. Solve the given case

Positive is spotted. TP=8, FP=2, FN=4, TN=6.

Build actual-row/predicted-column matrix and total.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Treat fern as positive. Counts are TP=6, FP=2, FN=3 and TN=9. How many examples and how many correct predictions are there?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 15 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Publish the Plant Classifier confusion matrix with positive-class definition, counts, denominator, and test-set version.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Positive=spotted. TP=6,FN=2,FP=3,TN=9. Draw a matrix with actual labels in rows and predictions in columns. Compute total and accuracy.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Build the count matrix

Use all eight worked IDs. Draw a matrix with reference rows and prediction columns, both ordered clear then blocked. Place the IDs in cells before writing the counts.

My reasoning and evidence

13. W2 Explain the two error types

Declare blocked positive. Give TP,TN,FP,FN counts and explain V3 and V4 in plain language. Does positive mean a desirable outcome?

My reasoning and evidence

14. W3 Audit totals and equal accuracy

Give the worked row totals, column totals, total count and accuracy. Compare its errors with $[[3,0],[3,2]]$ using the same axes and eight reference records.

My reasoning and evidence

15. W4 Count a fresh evaluation independently

Each pair below is reference then prediction: R1 clear/clear, R2 clear/blocked, R3 clear/blocked, R4 blocked/blocked, R5 blocked/clear, R6 blocked/blocked. Use clear then blocked on both axes and reference rows. Build the matrix, give row and column totals, name TP,TN,FP,FN with blocked positive, and calculate accuracy.

My reasoning and evidence

16. W5 Interpret a zero count carefully

A report gives zero false positives but omits the evaluated sample size, axes and class definitions. What information is missing, and why does zero not prove that future false alarms are impossible?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Confusion Matrix

Read the confusion matrix as four distinct stories about a plant decision.

Positive means spotted. Toy results: TP=4, FN=1, FP=3, TN=12. Rows are actual spotted/healthy; columns predicted spotted/healthy.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Positive means spotted

	predicted spotted	predicted healthy
actual spotted	4	1
actual healthy	3	12

Read a cell using both its row and column labels.

Confusion Matrix

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Fill actual spotted row

Actual spotted: [4, 1]. Four caught; one missed.



2. Fill actual healthy row

Actual healthy: [3, 12]. Three false alarms; twelve correctly clear.



3. Compute accuracy

Correct=4+12=16 of total 20, giving 80%.

CHANGE ONE THING

A review corrects one missed spotted leaf, moving it from FN to TP.

Explain the changed result using the matching lesson.

Why is adding TP alone without reducing FN an incorrect revision? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Read the confusion matrix as four distinct stories about a plant decision.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Correct=4+12=16 of total 20, giving80%.

1. Compute accuracy
2. Fill actual spotted row
3. Fill actual healthy row

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Compute accuracy Correct=4+12=16 of total 20, giving80%. Read the picture in order. Correct=4+12=16 of total 20, giving80%. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: A review corrects one missed spotted leaf, moving it from FN to TP. Which conclusion is supported by the supplied facts?

1. Counts become TP 5, FN 0, FP 3, TN 12; accuracy is $17/20=85\%$.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Counts become TP 5, FN 0, FP 3, TN 12; accuracy is $17/20=85\%$. One decision becomes correct while the total number of evaluated leaves remains 20.

Repair a missing building block

Review Scores and Calibration Intuition (ai-4/14). Goal: Interpret calibration using groups of forecasts. Short practice: Does a calibrated 0.8 score mean every such leaf is spotted? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Nearest Neighbors (ai-4/12). Goal: Compute absolute distances and rank training neighbours. Short practice: Why is the $k=1$ outcome at length 5 not uniquely determined without a tie rule? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Dataset Cards (ai-4/9). Goal: Inspect row meaning missing values and repeated records. Short practice: What two counts should the revised card show? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
